FIG. 15A — WRITE-AHEAD LOG — STRUCTURE & LIFECYCLE 1500

SECTION A: WRITE-AHEAD LOG STRUCTURE 1500

WAL File Layout (NDJSON) — {dataDir}/wal/partition-0.wal:

```
Line 1: {"entryId":"a3f2..","sequenceNumber":"0","type":"proof","partitionIndex":0,
          "walletId":"4829..","proofJson":"..","receiptData":{..},"auditEventData":{..},
          "stateSnapshot":{"registryRoot":"..","lifecycleRoot":"..","walletCount":"1",
          "auditChainHash":".."},"integrityHash":"b7c1..","timestamp":1709312000000,
          "committed":false}
Line 2: {"commit":"a3f2.."} <-- commit marker
Line 3: {"entryId":"d4e5..","sequenceNumber":"1","type":"proof",...,"committed":false}
Line 4: {"commit":"d4e5.."} <-- commit marker
Line 5: {"entryId":"f6a7..","sequenceNumber":"2","type":"batch_begin",...}
Line 6: {"entryId":"c8b9..","sequenceNumber":"3","type":"batch_commit",...}
Line 7: {"commit":"f6a7.."}
Line 8: {"commit":"c8b9.."} <-- crash here? lines 7-8 may not exist
```

WalEntry Fields: Six Entry Types:

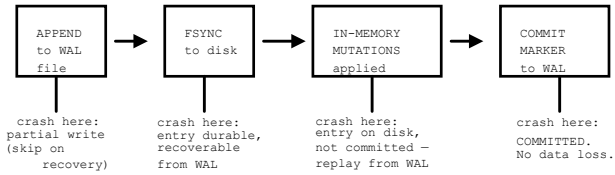
```
entryId: SHA-256 first 128 bits
   (type:part:seq:ts, truncated to 32 hex)
sequenceNumber: monotonic per partition
type: one of 6 types
partitionIndex: which partition
walletId?: target wallet
proofJson?: serialized ZK proof
receiptData?: {id, walletId, roots...}
auditEventData?: {walletId, roots...}
claimEntryData?: {leaf, metadata}
batchData?: {batchNumber, proofCount...}
integrityHash: SHA-256 first 128 bits
   (critical fields, truncated to 32 hex)
timestamp, committed, originNodeId?
```

```
'proof' ZK proof OK

'batch_begin' build start
'batch_commit' build done
'batch_abort' build failed
'settle' settlement OK
'fail' settle failed
```

```
stateSnapshot?: {registryRoot,
   lifecycleRoot, walletCount,

   auditChainHash}
```

SECTION B: WAL LIFECYCLE — APPEND -> COMMIT -> COMPACT 1502

WAL SAFETY CONTRACT:



COMPACTION (after checkpoint):

BEFORE: partition-0.wal AFTER: partition-0.wal

```
entry seq=0 (committed) <- remove
commit seq=0 <- remove
entry seq=1 (committed) <- remove
commit seq=1 <- remove
entry seq=2 (committed) <- remove
   <- checkpoint here
commit seq=2 <- remove
entry seq=3 (committed) <- KEEP
commit seq=3 <- KEEP
entry seq=4 (uncommitted) <- KEEP
```

```
entry seq=3 (committed)
commit seq=3
entry seq=4 (uncommitted)
```

```
truncateBefore(partition=0,
   beforeSeq=2):
1. Filter: keep seq > 2
2. Write to temp file
3. fsync temp
4. Atomic rename: temp -> .wal
```

Two implementations: FileWalProvider (production: NDJSON + fsync + atomic truncation + corrupt line skip) and InMemoryWalProvider (testing: Map<partition, WalEntry[]>, same IWalPro + ider API)